University of Trento
Master's Degree in Computer Science

Data Visualization
Technical Report

Data Visualization Lab
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Davide Donà	264467
Andrea Blushi	264468
Lorenzo Di Berardino	265984

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1 City Bike Dataset

For this project, our main dataset is the Citi Bike system dataset, a public repository tracking New York City bike trips since 2013.

1.1 Dataset Structure

Released monthly as open-data CSV files, these records log every completed ride, with high-volume months split across multiple files. Each row represents a single trip, offering detailed attributes that characterize the journey.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID**: A unique identifier for each bike trip.
- **Rideable Type**: The type of bike used for the trip ("classic" or "electric").
- **Started At**: A timestamp indicating when the trip started.
- **Ended At**: A timestamp indicating when the trip ended.
- **Start Info**: Information about the station and location where the trip started, structured as:
  - **Station Name**: The name of the starting station.
  - **Station ID**: The identifier of the starting station.
  - **Latitude**: The latitude coordinate of the starting station.
  - **Longitude**: The longitude coordinate of the starting station.
- **End Info**: Information about the destination station, structured equally to Start Info.
- **User Type**: The type of user ("member" or "casual").

1.1.2 Extracted Features

Additional per-trip features were derived to support the analysis:

- **Trip Duration**: The duration of the trip in seconds, calculated as Ended At $-$ Started At.
- **Trip Distance**: The distance between the start and end stations, computed using the Haversine formula and multiplied by a circuity factor of 1.3 to better approximate real routes.
- **Station Popularity**: For each station and time period, the number of trips starting and ending at that station, obtained by grouping trips by station.
- **Trips Flow**: For each pair of stations and time period, the number of trips travelling between them, obtained by grouping trips by station pair.

1.2 Analysis and Insights

For transparency and reproducibility, the complete workflow is available in our Jupyter notebook. Using the February 2025 trip dataset, the following sections examine temporal, spatial, and behavioral usage patterns. Rather than offering a comprehensive study of the Citi Bike system, this prior analysis serves to justify our visualization design choices.

Temporal Analysis Figure 1 illustrates distinct variations in demand across different hours and days of the week. Weekday trips peak during standard commuting hours, whereas weekend usage shifts toward leisure-oriented patterns.

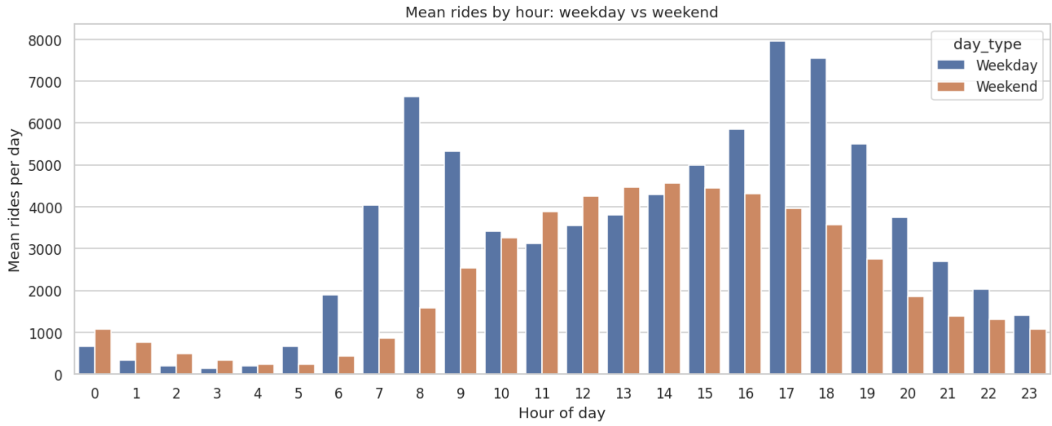

Figure 1: Distribution of trips by hour of the day and weekday/weekend.

Spatial Analysis Figure 2 highlights the distribution of station origins across the city. Station popularity and trip flows reveal that demand is concentrated around Manhattan and other high-density areas, with some stations acting as important origins.

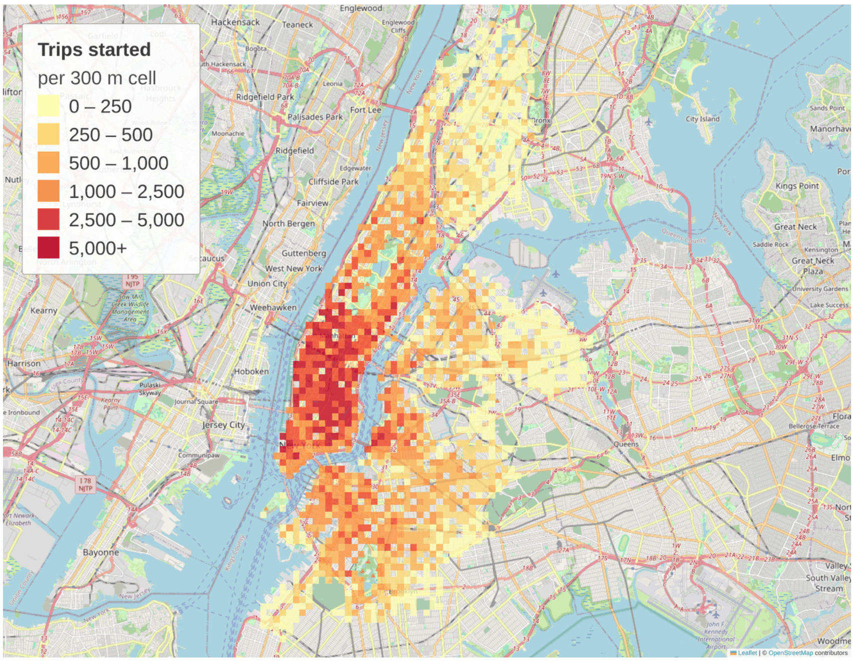

Figure 2: Distribution of trips by station of origin, aggregated into 300 m cells so that every colour class maps to an exact number of trips.

User/Bike Type Behaviour Analysis As shown in Figure 3, casual riders lean much more heavily on electric bikes than members do. Furthermore, casual riders take longer trips on both bike types, whereas members maintain a steady, short duration regardless of the bike they choose.

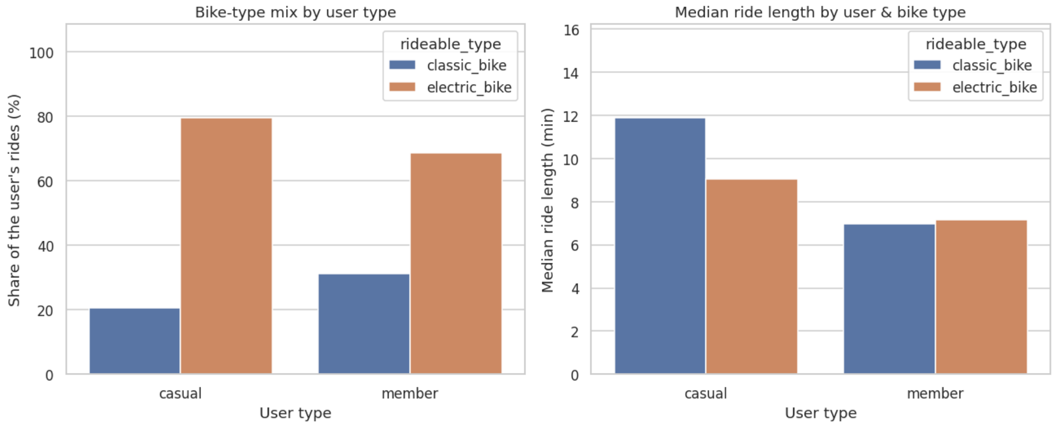

Figure 3: Distribution of trips by user/ride type based on rides and trip length.

Trip Duration and Distance Trip duration and distance distributions provide additional insight into mobility behaviour. Most trips are relatively short, while longer rides are less frequent and often correspond to recreational usage. It's important to note that durations begin at one minute, as Citi Bike removes trips shorter than 60 seconds at the source.

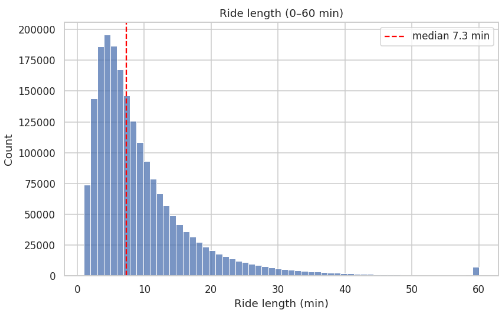

Figure 4: Distribution of trip durations.

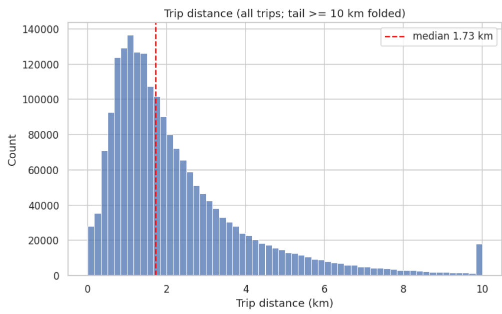

Figure 5: Distribution of trip straight-line distances.

1.3 Data Quality

To prevent misleading visual representations and maintain data integrity, we identified several data quality issues in the raw Citi Bike dataset prior to plotting.

Different Legacy Structure In 2020, Citi Bike updated its trip data schema, introducing column and formatting changes that differ from the 2013–2019 legacy format (detailed in our analysis notebook). Despite these structural discrepancies, the historical data remains invaluable. To integrate it seamlessly, our preprocessing pipeline standardizes all monthly files into a single, unified feature set.

Missing Station Values A tiny fraction of trips contain missing values, confined entirely to station-related fields. The issue predominantly impacts end-of-trip attributes: in the representative month, roughly 4,800 trips (0.24%) lack an end station identifier, compared to just 0.03% for start stations. This gap likely stems from abandoned bikes or off-dock returns. Because spatial and distance metrics require complete origin-destination pairs, these incomplete records are removed during preprocessing.

Trip Outliers Beyond missing values, a small number of trips are clear outliers. In a representative month, about 315 trips (0.02%) last up to 25 hours, appearing in the final bin of the duration histogram (Figure 4). These likely reflect unreturned bikes rather than continuous rides, so trips exceeding a maximum-duration threshold are dropped. Additionally, around 27,000 trips (1.4%) start and end at the same station, resulting in a zero straight-line distance (Figure 5); because distance metrics are meaningless for these round trips, they are removed. Finally, some trips report impossible distances (11,000 km as an example), due to corrupted station coordinates or maintenance trips, and are filtered out using a maximum-distance threshold.

Biases This dataset carries certain inherent limitations. Because it relies exclusively on the Citi Bike system dataset, the findings may not accurately reflect the demographics of the entire population of cyclists in New York City. Additionally, trip distances are calculated using the Haversine formula between the start and end stations; this straight-line metric serves only as a geometric approximation and cannot account for the actual routes or detours taken by riders. Unfortunately, these structural data constraints cannot be resolved within the scope of our project.

2 Station Metadata Dataset (GBFS)

To enrich the static station data provided by Citi Bike, we incorporate their real-time feed, available on the Citi Bike System Data portal. This live feed offers direct insight into the current capacity, operational status, and physical infrastructure of individual stations.

2.1 Dataset Structure

Unlike historical trip records, station metadata is served via a public JSON API adhering to MobilityData's GBFS standard. From this service, we query two key Citi Bike endpoints:

- `station_information.json`: Static properties like location and capacity, whose content changes only when the network itself changes.
- `station_status.json`: Live availability and operational state.

2.1.1 Original Structure

The two endpoints share the `station_id` field, on which they are merged to produce a unified station record comprising:

- **From** `station_information`:
  - **Station ID**: Internal GBFS identifier, a UUID for most stations and a numeric string for the rest.
  - **Short Name**: Public station identifier used for joins with Citi Bike trip data.

  - **Name**: Station name.
  - **Latitude** and **Longitude**: Station coordinates.
  - **Capacity**: Total number of docks installed at the station.

- **From** `station_status`:
  - **Bikes Available**: Currently rentable bikes.
  - **Vehicle Types Available**: Per-type counts of the bikes currently at the station.
  - **Docks Available**: Currently available docks.
  - **Operational Flags**: `is_installed`, `is_renting`, and `is_returning`.

2.1.2 Extracted Features

Using the three operational flags (`is_installed`, `is_renting`, and `is_returning`), we derive a boolean `active` indicator to filter out unavailable stations from live views. Evaluating all three status flags is necessary because physical installation does not imply service: stations can remain installed while suspending both rentals and returns. Relying solely on `is_installed` would therefore misclassify inactive stations as available.

2.2 Analysis and Insights

This section examines the structural and operational characteristics of the Citi Bike network, providing a foundational snapshot of its system infrastructure. Because the feed is a dynamic snapshot, this analysis represents a single point in time rather than a strictly reproducible state.

Capacity Distribution Figure 6 illustrates network capacity distribution, highlighting the contrast between smaller local stations and major transit hubs. Notably, the count at 0 docks does not indicate corrupted data; rather, it represents 42 inactive stations currently present in the feed (as detailed in our Jupyter notebook).

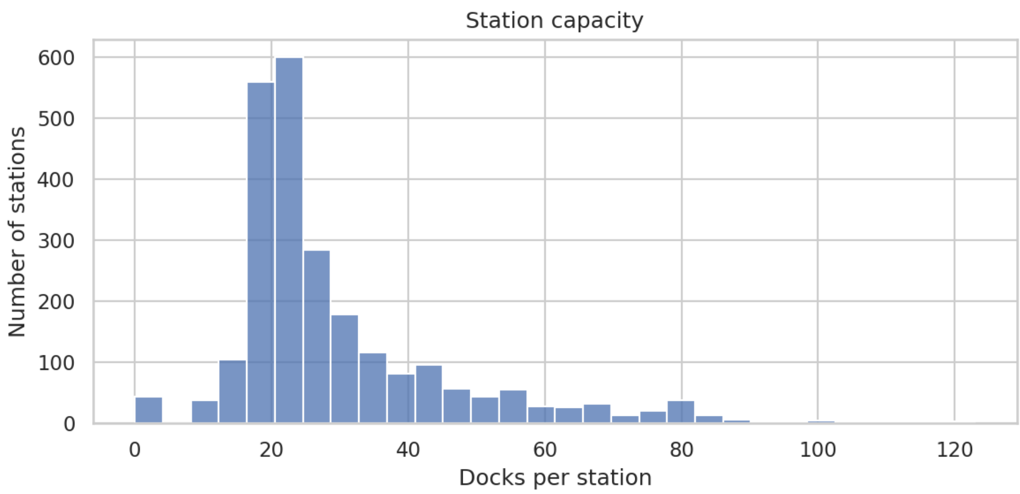

Figure 6: Distribution of station capacity, measured as the number of docks per station.

Live Availability Live bike and dock availability reflects the real-time operational state of the system. Figure 7 decomposes total network capacity by classifying every dock into one of six mutually exclusive states, with separate breakdowns for active and inactive stations to provide a more granular view. Since those breakdowns are shares computed within each group, the left panel first sizes the groups themselves, in stations and in docks, so that the inactive share is not read as larger than it is. Delivering these status classifications dynamically at the station level is precisely what makes this live feed a valuable component of the interactive dashboard.

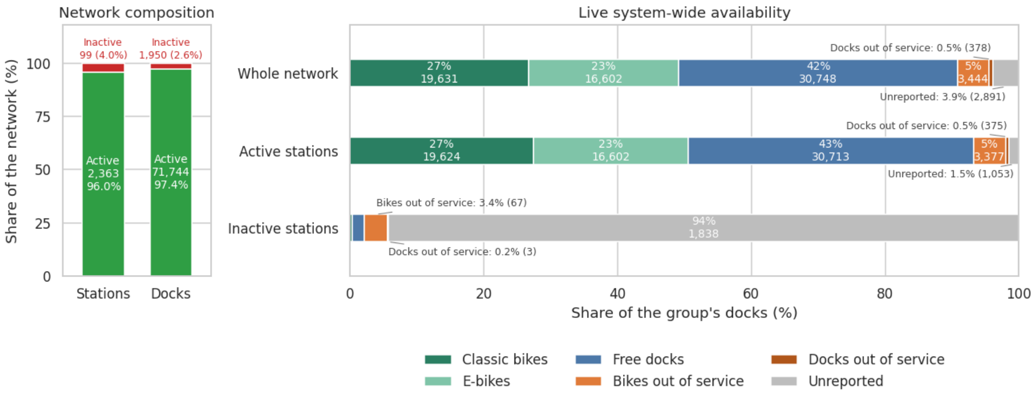

Figure 7: System-wide capacity breakdown across all, active, and inactive stations. The left panel gives the size of each group, in stations and in docks. In the right panel, total capacity is fully decomposed into six mutually exclusive states: rentable classic bikes, rentable e-bikes, free docks, out-of-service bikes, out-of-service docks, and unreported capacity.

2.3 Data Quality

To maintain data integrity and prevent visual distortion, our initial validation focused on three core quality issues: unreported capacity, feed self-consistency, and identifier alignment. Ultimately, as confirmed in our analysis notebook, the live feed contains no missing station values.

Inactive Stations Stations undergoing installation, repair, or decommission are grouped under the `inactive` category, representing 99 out of 2,462 stations (4.0%) and 1,950 docks (2.6% of total capacity). As illustrated in Figure 7, these stations declare capacity while broadcasting almost no live counts, and the few bikes assigned to them are nearly all out of service. Retaining them would distort overall capacity metrics and render ghost (false empty) stations on the interactive map without contributing usable availability.

Unreported Capacity The unreported capacity (Figure 7) arises from using two different endpoints. Station capacity is a static property provided by `station_information` and updated periodically, whereas live bike and dock counts are delivered continuously by `station_status`. Filtering out the inactive stations described above eliminates the main source of variance without altering real availability. For any remaining active stations that do not reconcile, dock states are read directly from live counters, treating capacity as a rough benchmark.

Identifier Consistency GBFS identifies stations in two ways: an internal `station_id` (either a UUID or a numeric string) and a public `short_name`. Trip history files use only the latter: 93.8% of station references in historical trips match a `short_name`, whereas none match a `station_id`. Therefore, all joins between live metadata and trip history rely on the `short_name` field.

Live-Only Metadata Citi Bike provides station metadata only as a live feed, with no historical archives. Consequently, removed stations retain their past trip records but lose all feed attributes, while newly added stations lack historical data, making them appear prematurely inactive. This bias only affects feed-specific attributes like capacity and operational status, as each trip record independently contains its own station name and coordinates.

3 Weather Dataset

To enrich the dashboard with meteorological analysis, we incorporate hourly NYC weather data retrieved from the Open-Meteo archive API.

3.1 Dataset Structure

Weather data are queried via the Open-Meteo JSON API by specifying coordinates, a date range, and variables. The API returns parallel hourly arrays derived from reanalysis models that update with a few days' delay, keeping the historical analysis stable in our case.

3.1.1 Original Structure

The dataset includes several attributes, but we consider only the following main fields:

- **Timestamp**: Hourly observation time.
- **Temperature**: Air temperature in °C.
- **Precipitation**: Hourly precipitation in millimetres.
- **Weather Code**: WMO weather condition code.
- **Wind Speed**: Wind speed in km/h.

Weather timestamps are downloaded in UTC and converted to local time (`America/New_York`) so they match the Citi Bike trip data. This is done because requesting local time directly applies a single time offset to the whole period, which creates a one-hour error whenever daylight saving time changes. To prevent missing data after adjusting for local time, we add a one-day buffer to both ends of our search window and trim it back to the exact target dates after conversion.

3.2 Analysis and Insights

Weather conditions serve as a key driver of daily bike usage. Rather than examining weather in isolation, the following section focuses directly on its relationship with overall ridership patterns.

Ridership vs. Weather Figure 8 illustrates daily ridership alongside mean temperature and total precipitation. The data shows a clear inverse relationship between precipitation and usage: the month's peak precipitation directly coincides with its lowest ridership volume. Meanwhile for the temperature, it's possible to see also a positive correlation with ridership, where the rides follows the temperature increase. This confirms that weather tracking provides valuable context for interpreting demand fluctuations and supporting operational planning.

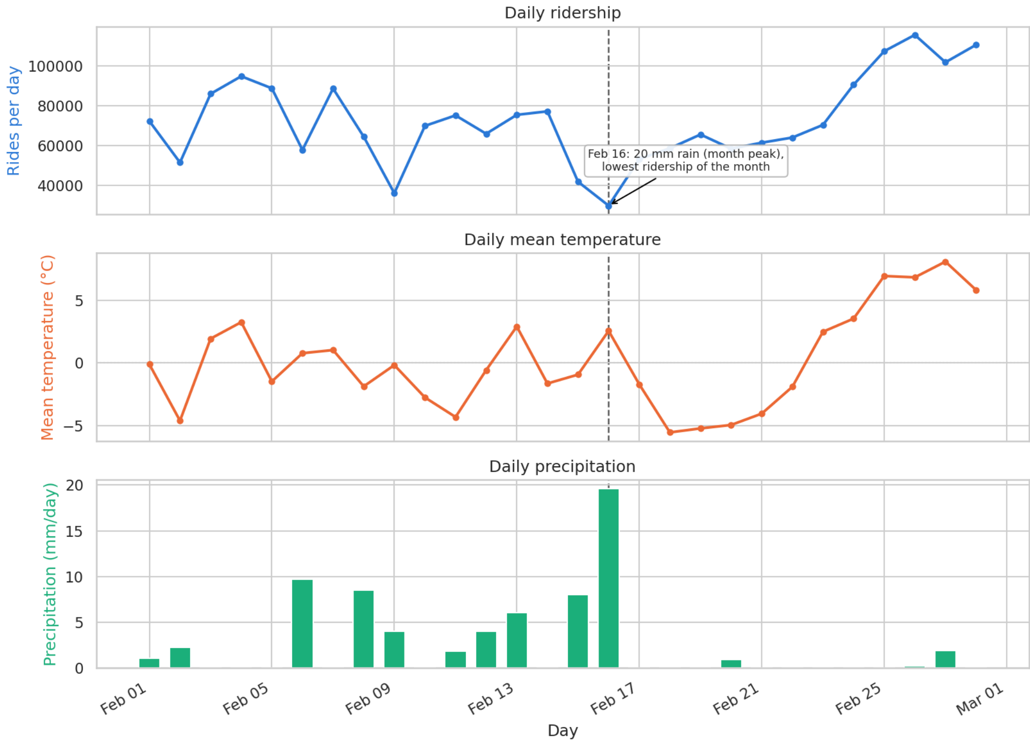

Figure 8: Daily ridership, mean temperature, and total precipitation over the reference month, plotted on aligned date axes with individual scales. The dashed guide highlights 16 February, the month's wettest day and also its least ridden.

3.3 Data Quality

The archive is complete, with an hourly series covering the entire reference year and no missing values in any variable. The issues we found therefore concern how the values are localised and what they represent, rather than gaps in the data.

Spatial Coverage Weather data is retrieved for a single point centered on Central Park and applied across the entire system. While this dataset misses microclimate variations across a more granular spatial scale, applying higher spatial granularity would introduce more modeling challenges than necessary for a city-scale analysis.

4 Bike Lanes Dataset

To enrich our station and trip insights, we incorporate New York City's bike-route network, retrieved from the NYC OpenData portal.

4.1 Dataset Structure

Distributed as a single CSV file, the dataset provides a current snapshot of the bike network alongside a historical record, with the NYC Department of Transportation updating the file on an annual basis.

4.1.1 Original Structure

Each record represents a single segment of a bike facility, with the following fields:

- **Geometry**: The set of lines defined in coordinates tracing the bike route.
- **Street**: The street the facility runs along.
- **From Street**, **To Street**: The cross streets that define the segment's endpoints.
- **Borough**: Which of the five boroughs the segment lies in.
- **Facility Class**: The type of facility across four classes:
  - **Protected**: Separated from motor traffic by physical barriers (curbs, bollards, greenways).
  - **Conventional**: Marked by painted lanes only, without physical barriers.
  - **Shared / Signed**: Standard travel lanes with signage or sharrows indicating bicycle traffic.
  - **Link**: Non-bikeable connectors (stairs, walkways) bridging gaps in the network.
- **Status**: Whether the facility is still in service or has been retired/modified.
- **Installation Date** and **Retirement Date**: When the facility opened and, for facilities no longer in service, when it was removed or superseded.
- **Route ID**: The named route the segment forms part of.
- **Segment ID**: Which piece of street the record covers. The numbering comes from LION (*Linear Integrated Ordered Network*), the city's official street map, so every facility is pinned to a known piece of road.

Although finer classifications are recorded for each facility, they are excluded during ingestion. Because these sub-types aggregate naturally into four primary classes, omitting them preserves structural simplicity without affecting anything the visualization shows.

4.2 Analysis and Insights

The analysis provides a spatial and temporal perspective on the bike lane network.

Spatial Overview Figure 9 maps the active network by facility class. Its spatial distribution closely mirrors the high-demand trip areas shown in Figure 2. This alignment confirms the facility data's reliability, laying a solid foundation for integration with complementary datasets.

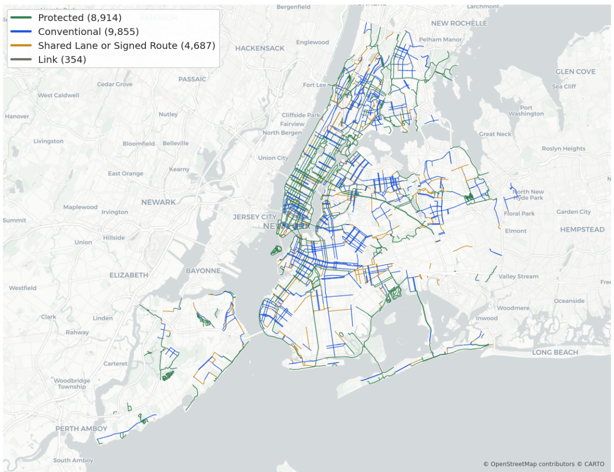

Figure 9: The bike-route network in service, colouring segments by facility class.

Temporal Overview Retaining retired facilities allows us to reconstruct the network's evolution year by year. Figure 10 displays annual installations above the horizontal axis and retirements below it, highlighting rapid network growth in recent years.

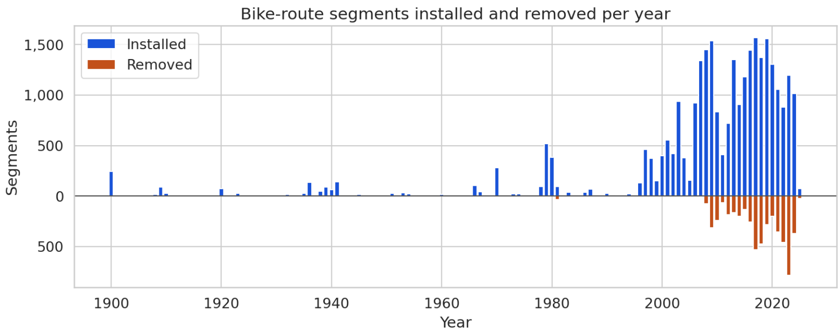

Figure 10: Segments installed and removed per year.

4.3 Data Quality

Apart from the retirement date, which is empty by design for facilities still in service, every column we consume is fully populated, every geometry parses, and none falls outside New York, all verified in the Jupyter notebook.

Sentinel Installation Dates Although the installation date field is never null, `01/01/1900` acts as a floor for dates previous to the 1900s (Figure 10). It appears in 244 segments (0.8%), 196 of which remain active, concentrated on long-standing park and parkway paths such as Ocean Parkway and Eastern Parkway whose routes predate the city's record-keeping.

Facility Classification Bias Facility class reflects the highest class present along a segment, as reported in the documentation. For instance, a street with a protected lane on one side and a shared lane on the other is classified entirely as protected. This affects 720 segments (2.5%): a minor share that barely alters the map's overall coloring, but implies that segment classes represent the best available facility rather than conditions on both sides.